

Epistemic Sybil Resistance: Information Aggregation Under Replicated Evidence

Marc Bara

Working paper, August 2026

Abstract

Collective inference systems often treat agreement among multiple reports as additional evidence. This is justified when reports contain sufficiently independent information, but it fails when apparent corroboration is generated by repeated transformations of shared evidence. We formalize this failure as an epistemic Sybil problem. Given existing reports R about a latent state Θ , an additional report Z is an epistemic Sybil extension when

$$I(\Theta; Z \mid R) = 0.$$

The definition concerns marginal information rather than identity, textual similarity, or source count. We show first that no report-only aggregator can generally distinguish redundant replication from independent corroboration: identical observable reports can arise under information structures requiring different Bayesian posteriors. We then establish a no-minting result from the data-processing inequality: arbitrarily many descendants of a fixed evidence set cannot contain more information about Θ than their evidential ancestors. Common ancestry, however, does not imply complete redundancy. In a Gaussian shared-root model, repeated noisy extractions provide positive but diminishing information and converge to a source-level ceiling. The same model yields exactly the classical effective-sample-size discount $1/[1 + \rho(m - 1)]$ under equicorrelation. A Monte Carlo study illustrates the resulting calibration failure: when 32 reports derive from one primitive root, naive independent pooling reduces nominal 95% posterior coverage to 38.1%, while dependence-aware Bayesian aggregation retains approximately 95% coverage. These results motivate aggregation relative to explicit side-information interfaces, such as authenticated provenance, rather than universal report deduplication.

1 Introduction

Suppose three analysts predict that a firm will default. One has examined its audited accounts, another has access to payment data from its banks, and a third has independently interviewed major suppliers. Now consider three other analysts who produce different and detailed arguments but ultimately derive them from the same credit-rating report.

Both situations contain three reports and may generate the same vote or confidence level. They nevertheless represent different information structures.

The distinction matters increasingly because generating reports has become cheap. A source can be summarized, translated, paraphrased, retrieved into multiple contexts, processed by different models, transmitted between agents, and returned to an aggregator in forms that appear unrelated. Conversely, genuinely independent evidence may produce near-identical reports. Observable diversity and evidential diversity therefore need not coincide.

The underlying statistical problem is old. Dependent probability forecasts cannot in general be combined as if their information sources were independent (Winkler, 1981; Clemen and Winkler, 1999). Distributed sensor fusion has long studied the repeated use of information,

including data incest, in which information re-enters a fusion process through network paths or loops (McLaughlin, Evans, and Krishnamurthy, 2003). Covariance Intersection provides conservative fusion when relevant cross-correlations are unknown (Julier and Uhlmann, 1997). Social-learning models similarly show that repeated communication can produce excessive influence when decision makers neglect repetition or correlation (DeMarzo, Vayanos, and Zwiebel, 2003; Golub and Jackson, 2010; Enke and Zimmermann, 2019).

Recent AI systems instantiate the same structural problem in new forms. Yan et al. (2026) show that misinformation originating with one LLM agent can propagate through other agents and acquire apparent collective support; their experiments explicitly trace evidence-root lineage. Berriaud and Wattenhofer (2026) construct weighting rules robust to duplicates and near-duplicates in metric spaces. Lee and Kim (2026) demonstrate a complementary failure in retrieval-augmented generation: lexically diverse passages can manufacture corroboration while evading near-duplicate filters. Wu (2026) explicitly groups sources by common upstream origin and discounts dependent evidence. Louck (2026) uses origin binding and Sybil-resistant corroboration to prevent manufactured agreement from elevating poisoned agent memories.

These literatures establish that repeated or dependent evidence can be overweighted. The objective of this paper is narrower: to identify what property should remain invariant when apparent corroboration is multiplied, and what an aggregator must know in order to enforce that invariance.

Let Θ denote a latent state and let

$$R = (R_1, \dots, R_n)$$

denote reports already admitted by an inference system. We call an additional report Z an epistemic Sybil extension when

$$I(\Theta; Z \mid R) = 0.$$

The report may have a different identity, use different language, present a different argument, or pass through an independent computational process. The defining property is that, conditional on what has already been admitted, it contributes no additional information about Θ .

This reframes Sybil resistance. Classical Sybil attacks multiply identities. An epistemic Sybil multiplies apparent evidential support without multiplying information about the state. The phenomenon can be adversarial, but it can also arise through honest retransmission.

The paper makes five contributions.

First, it defines epistemic replication using conditional mutual information rather than report identity or similarity.

Second, it shows that epistemic Sybil resistance must be defined relative to the side information available to an aggregator. With no information about the generating structure, redundant repetition and independent corroboration are generally not identifiable from reports alone.

Third, the data-processing inequality gives a source-level no-minting principle. Processing a fixed evidence set may reveal information already latent in that evidence, but multiplying its descendants cannot create additional information about the state beyond what their ancestors contain.

Fourth, common ancestry does not justify counting evidence roots as indivisible units. A Gaussian shared-root model shows that multiple extractions from one source can add information, but with diminishing returns toward a finite ceiling. Under equicorrelation, the same model yields the standard effective-sample-size factor exactly.

Fifth, a controlled Monte Carlo study quantifies the practical consequence. When report count rises while primitive evidence remains fixed, an independence-assuming aggregator becomes increasingly overconfident even though the underlying information approaches a ceiling.

Together, the results distinguish report multiplicity, source multiplicity, and information multiplicity.

2 Related Work

2.1 Dependent forecasts and information aggregation

Dependence among information sources has long been recognized as central to forecast combination. Winkler (1981) studies the combination of probability distributions derived from dependent information sources, while Clemen and Winkler (1999, 2007) review dependence and calibration issues in expert aggregation.

This literature establishes an important novelty boundary. The present paper does not claim that correcting for correlated evidence is new. Its focus is the information structure created when one evidential origin can be transformed into arbitrarily many heterogeneous reports and the dependence needed for correct aggregation is not observable from those reports themselves.

2.2 Data incest and covariance intersection

Distributed estimation provides a close precursor. When estimates are repeatedly exchanged across a network, information can return to an estimator embedded in another estimate. Treating the returned estimate as independent counts some observations repeatedly and leads to overconfident uncertainty estimates.

Covariance Intersection addresses a related problem when cross-correlations are unknown. Recent work on Overlapping Covariance Intersection extends this idea to settings in which partial structural knowledge of shared correlations is available (Pedroso, Batista, and Heemels, 2026).

These methods provide natural candidates for conservative aggregation when provenance reveals possible dependence but does not identify a complete probabilistic model. Section 9 returns to this connection.

2.3 Social learning and correlation neglect

Economics provides a behavioral counterpart. DeMarzo, Vayanos, and Zwiebel (2003) show how repeated transmission can create influence unrelated to informational accuracy. Golub and Jackson (2010) characterize conditions under which naive social learning nevertheless aggregates information successfully. Enke and Zimmermann (2019) provide experimental evidence that people systematically neglect correlation and double count information derived from common sources.

An epistemic Sybil can be interpreted as a strategic or algorithmic version of this phenomenon. Generative systems sharply reduce the cost of retelling while simultaneously making common ancestry harder to infer from surface form.

2.4 Classical Sybils and false-name manipulation

Douceur (2002) introduced the classical Sybil attack as the creation of multiple identities by one participant. False-name manipulation in mechanism design studies the corresponding possibility that one economic actor presents itself as multiple participants.

Burnat and Davidson (2026) recently apply this perspective to machine-learning data attribution. Their quotient mechanisms aggregate over evidence-backed attribution clusters rather than raw identities, preventing contributors from increasing attributed value merely by splitting or duplicating contributions.

The protected resource here is different. Epistemic Sybil resistance concerns neither identity weight nor payment attribution, but marginal information relevant to a belief.

2.5 Clone-robust aggregation

Berriaud and Wattenhofer (2026) formalize clone robustness for arbitrary metric spaces. Their mechanisms prevent exact or near-duplicate reports from obtaining additional aggregate influence merely through multiplication.

Metric clone robustness and epistemic Sybil resistance are related but non-equivalent. Identical reports can result from independent observations, while distant reports can be transformations of a single observation. Theorem 1 formalizes why report-space representation alone cannot universally distinguish these cases.

2.6 Provenance-aware aggregation

Recent AI research increasingly records evidence lineage. Yan et al. (2026) trace evidence roots while studying misinformation propagation in multi-agent systems. Xu (2026) uses content-addressed lineage as a global side channel in decentralized mesh inference. Louck (2026) binds authority to origin and preserves it through subsequent transformations.

Wu (2026) is particularly close to the present statistical problem. Sources sharing an upstream origin are placed in dependence blocks, and evidence from a block of size m is discounted using

$$\kappa_m = \frac{1}{1 + \rho(m - 1)}.$$

Wu treats the expression as an effective-sample-size approximation. Section 6 shows that it is exact within a homoscedastic Gaussian equicorrelation model, giving the coefficient a direct generative interpretation.

2.7 Blackwell ordering and partial information

The data-processing argument used below has a decision-theoretic counterpart in Blackwell’s comparison of experiments. If

$$\Theta \rightarrow E \rightarrow Y,$$

then Y is a garbling of E . Observing E therefore Blackwell-dominates observing Y : no decision maker can obtain lower Bayes risk from the garbled observation for any decision problem satisfying the standard conditions (Blackwell, 1953).

Partial Information Decomposition supplies another interpretation. In a nonnegative two-source PID, conditional mutual information can be written as the sum of information unique to Z and information available only synergistically from R and Z . Therefore, if

$$I(\Theta; Z \mid R) = 0,$$

an epistemic Sybil contributes neither unique nor synergistic information about Θ beyond R (Williams and Beer, 2010).

2.8 Information elicitation

Peer-prediction mechanisms seek truthful reports when direct verification of the underlying truth is unavailable. The Mutual Information Paradigm of Kong and Schoenebeck (2016) uses information-theoretic dependence between peer reports to reward informative behavior.

This creates a tension. Correlation between reports can arise because independent signals are informative about a common state, or because one report derives from another. A

mechanism that rewards agreement without identifying the causal origin of that agreement can reward endogenous dependence. Truthful provenance is therefore a distinct mechanism-design problem from truthful content elicitation.

3 Model

Let Θ be a latent random variable on state space $\mathcal{T}$. An inference system receives

$$R = (R_1, \dots, R_n)$$

and produces a belief

$$A(R) \in \Delta(\mathcal{T}).$$

A report may originate from a sensor, human, model, database, document, retrieval operation, algorithm, or any composition of them.

3.1 Independent corroboration

Two reports X and Z satisfy the standard conditional-independence model when

$$X \perp Z \mid \Theta.$$

Agreement between them can then add substantial evidence.

3.2 Shared-origin evidence

Two reports can instead derive from a common evidence variable E :

$$\Theta \rightarrow E \rightarrow (X, Z).$$

Common ancestry induces dependence, but does not imply that one report is redundant conditional on the other. Different reports can extract different aspects of the same evidence.

3.3 Zero-novelty evidence

The limiting form of redundancy is

$$\Theta \perp Z \mid R,$$

equivalently,

$$I(\Theta; Z \mid R) = 0.$$

Definition 1 (Epistemic Sybil extension). Given an admitted report profile R , an additional report Z is an epistemic Sybil extension relative to Θ if

$$I(\Theta; Z \mid R) = 0.$$

A collection $Z_{1:k}$ is an epistemic Sybil extension if

$$I(\Theta; Z_{1:k} \mid R) = 0.$$

The definition is invariant to representation. A Sybil extension need not duplicate an existing string, embedding, identity, argument, or source label.

Define its incremental epistemic contribution as

$$\Gamma(Z; R) = I(\Theta; Z \mid R).$$

Thus,

$$\Gamma(Z; R) = 0$$

for an epistemic Sybil extension, while

$$\Gamma(Z; R) > 0$$

for a report containing marginal information about Θ .

The quantity is a theoretical target. It is not assumed directly observable.

3.4 Side-information interfaces

Epistemic Sybil resistance cannot be defined independently of what the aggregator is allowed to observe.

Let $\mathcal{P}$ be a class of information-generating models. A side-information interface $\mathcal{J}$ maps model $P \in \mathcal{P}$ and an observed report profile R to side information

$$S = \mathcal{J}_P(R)$$

available to the aggregation mechanism.

Examples include

$\mathcal{J}_0$: no structural side information,

$\mathcal{J}_G$: authenticated provenance information,

and

$\mathcal{J}_*$: the complete relevant joint information structure.

An aggregation mechanism relative to $\mathcal{J}$ is written

$$A_{\mathcal{J}}(R, \mathcal{J}_P(R)).$$

Definition 2 (Epistemic Sybil resistance relative to an interface). An aggregation mechanism $A_{\mathcal{J}}$ is epistemic-Sybil-resistant on model class $\mathcal{P}$ relative to side-information interface $\mathcal{J}$ if, whenever $P \in \mathcal{P}$ and an extension Z satisfies

$$I_P(\Theta; Z \mid R) = 0,$$

then

$$A_{\mathcal{J}}(R, Z, \mathcal{J}_P(R, Z)) = A_{\mathcal{J}}(R, \mathcal{J}_P(R))$$

almost surely.

Resistance is therefore always relative to an information interface. Section 5 establishes a limitation under $\mathcal{J}_0$; Sections 7 through 9 examine what richer interfaces provide.

The defining conditional-independence relation also has a sufficiency interpretation. If

$$I(\Theta; Z \mid R) = 0,$$

then R remains sufficient for Θ after the observation space is enlarged from R to (R, Z) .

4 The No-Minting Principle

Let E represent primitive evidence and let

$$Y = (Y_1, \dots, Y_m)$$

be any collection of reports generated solely from E . Assume

$$\Theta \rightarrow E \rightarrow Y.$$

Proposition 1 (Evidence no-minting).

$$I(\Theta; Y) \leq I(\Theta; E).$$

Proof. The result follows directly from the data-processing inequality. $\square$

The proposition gives a source-level conservation principle: processing can expose information already contained in evidence, but multiplying its descendants cannot create information about the underlying state that was absent from their ancestry.

The corresponding Blackwell result is stronger than the Shannon statement. Since Y is a garbling of E , observing E is at least as useful as observing Y for every Bayesian decision problem under the standard comparison-of-experiments framework.

Importantly, Proposition 1 does not imply that every descendant of E after the first should be discarded. A first report can reveal only part of the information latent in E . Additional processing can expose more. The total information in all descendants remains bounded by the information present in their ancestry.

This distinction separates exact replication from repeated extraction.

5 Report-Only Non-Identifiability

Let

$$\Theta \sim \text{Bernoulli}(1/2),$$

and let a binary report X satisfy

$$P(X = \Theta) = p, \quad p \in (1/2, 1).$$

Consider two possible information structures.

5.1 Clone structure

The second report is an exact copy:

$$Z = X.$$

Upon observing $X = Z = 1$,

$$P_C(\Theta = 1 \mid X = 1, Z = 1) = p.$$

5.2 Independent structure

Instead suppose

$$X \perp Z \mid \Theta$$

and both reports have accuracy p . Then

$$P_I(\Theta = 1 \mid X = 1, Z = 1) = \frac{p^2}{p^2 + (1 - p)^2}.$$

For $p > 1/2$,

$$\frac{p^2}{p^2 + (1 - p)^2} > p.$$

The observable report profile is identical in the two structures, while the appropriate posterior differs.

Theorem 1 (Report-only non-identifiability). *Consider an aggregation rule operating under $\mathcal{J}_0$ and observing only (X, Z) together with the common marginal accuracy p . No deterministic rule can be Bayes-correct under both the clone and independent information structures.*

If its posterior estimate after observing $(1, 1)$ is q , then

$$\max \left\{ |q - p|, \left| q - \frac{p^2}{p^2 + (1 - p)^2} \right| \right\} \geq \delta(p),$$

where

$$\delta(p) = \frac{p(1 - p)(2p - 1)}{2[p^2 + (1 - p)^2]}.$$

The same lower bound holds for randomized rules in expected absolute error.

Proof. The Bayes-optimal posterior values in the two structures are

$$q_C = p$$

and

$$q_I = \frac{p^2}{p^2 + (1 - p)^2}.$$

Because the observable input is the same, a deterministic mechanism must choose the same value q in both cases. Therefore,

$$\max\{|q - q_C|, |q - q_I|\} \geq \frac{|q_I - q_C|}{2}.$$

Since

$$q_I - q_C = \frac{p(1 - p)(2p - 1)}{p^2 + (1 - p)^2},$$

the deterministic result follows.

For a randomized output Q ,

$$\mathbb{E}|Q - q_C| + \mathbb{E}|Q - q_I| \geq |q_I - q_C|,$$

so at least one expected absolute error must be at least half the posterior separation. $\square$

The mechanism in Theorem 1 is deliberately granted knowledge of p . Withholding the marginal reliability would only make identification more difficult.

For $p = 0.7$,

$$q_C = 0.700, \quad q_I \approx 0.845,$$

and therefore

$$\delta(0.7) \approx 0.072.$$

Corollary 1 (Incompatibility under the trivial interface). *Over a model class containing both clone and independent-corroboration structures, no rule operating under $\mathcal{J}_0$ can guarantee both epistemic-Sybil invariance and Bayes-correct responsiveness to independent corroboration.*

The result does not imply that semantic similarity is useless. Similarity can be statistically informative about common ancestry. It cannot, however, provide a universally valid identification criterion.

6 Shared Roots and Effective Evidence

The opposite error is to conclude that every set of reports sharing one evidence root should count as exactly one observation.

Suppose

$$E = \Theta + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2),$$

and m reports independently extract information from E :

$$R_i = E + \eta_i, \quad \eta_i \sim \mathcal{N}(0, \nu^2).$$

Initially assume that the η_i are mutually independent and independent of ε and Θ .

Conditional on Θ ,

$$R \sim \mathcal{N}(\Theta \mathbf{1}, \Sigma_m),$$

with

$$\Sigma_m = \sigma^2 \mathbf{1} \mathbf{1}^\top + \nu^2 I.$$

The likelihood precision about Θ is

$$J_m = \mathbf{1}^\top \Sigma_m^{-1} \mathbf{1}.$$

Proposition 2 (Shared-root saturation). *For $\nu^2 > 0$,*

$$J_m = \frac{m}{\nu^2 + m\sigma^2} = \frac{1}{\sigma^2 + \nu^2/m}.$$

Hence

$$\lim_{m \rightarrow \infty} J_m = \frac{1}{\sigma^2}.$$

The marginal precision generated by the m -th report is

$$J_m - J_{m-1} = \frac{\nu^2}{(\nu^2 + m\sigma^2)[\nu^2 + (m-1)\sigma^2]},$$

which is positive and of order $O(m^{-2})$.

Proof. By the Sherman–Morrison identity,

$$\Sigma_m^{-1} = \frac{1}{\nu^2} I - \frac{\sigma^2}{\nu^2(\nu^2 + m\sigma^2)} \mathbf{1} \mathbf{1}^\top.$$

Consequently,

$$J_m = \mathbf{1}^\top \Sigma_m^{-1} \mathbf{1} = \frac{m}{\nu^2 + m\sigma^2}.$$

The remaining expressions follow directly. □

6.1 Three scaling regimes

The model separates three cases.

For exact replication, $\nu^2 = 0$, so every report is the same observation E and

$$J_m = \frac{1}{\sigma^2}.$$

Repetition adds no information.

For repeated extraction with $\nu^2 > 0$, additional reports reduce extraction noise. Their contribution is positive but saturates because the common error ε cannot be averaged away.

If each report instead has an independent root,

$$E_i = \Theta + \varepsilon_i,$$

with independent ε_i , then

$$J_m^{\text{ind}} = \frac{m}{\sigma^2 + \nu^2}.$$

Precision now grows linearly with report count.

Thus,

exact replication, shared-root extraction, independent corroboration

have fundamentally different information scaling.

6.2 Exact effective-sample-size discount

Let

$$v = \sigma^2 + \nu^2$$

be the conditional variance of an individual report. The intraclass correlation conditional on Θ is

$$\rho = \frac{\sigma^2}{\sigma^2 + \nu^2}.$$

Since

$$\nu^2 + m\sigma^2 = v[1 + \rho(m - 1)],$$

Proposition 2 can be written as

$$J_m = \frac{m}{1 + \rho(m - 1)} \frac{1}{v}.$$

Define

$$m_{\text{eff}} = \frac{m}{1 + \rho(m - 1)}.$$

The m correlated reports therefore contain exactly the same likelihood precision as m_{eff} independent reports with variance v .

Corollary 2 (Exact effective-sample-size discount under Gaussian equicorrelation). *Within the shared-root Gaussian model,*

$$\kappa_m = \frac{1}{1 + \rho(m - 1)}$$

is exactly

$$\frac{m_{\text{eff}}}{m}.$$

The dependence discount used by Wu (2026) therefore coincides with the exact precision correction implied by this model. The result is specific to the homoscedastic Gaussian equicorrelation class; applying the same coefficient to arbitrary evidence structures is not generally Bayes-optimal.

Within the model, ρ has a direct interpretation: it is the proportion of each report's conditional variance attributable to the common source component.

6.3 Correlated extraction

Independence among extraction errors can be unrealistic when several AI agents share the same base model.

Suppose

$$\text{Var}(\eta_i) = \nu^2$$

and for $i \neq j$,

$$\text{Cov}(\eta_i, \eta_j) = \gamma\nu^2, \quad 0 \leq \gamma < 1.$$

The total conditional variance remains

$$v = \sigma^2 + \nu^2,$$

while off-diagonal covariance becomes

$$c = \sigma^2 + \gamma\nu^2.$$

The resulting intraclass correlation is

$$\rho_{\text{tot}} = \frac{\sigma^2 + \gamma\nu^2}{\sigma^2 + \nu^2}.$$

Hence,

$$J_m = \frac{m}{v[1 + \rho_{\text{tot}}(m - 1)]},$$

and when $c > 0$,

$$\lim_{m \rightarrow \infty} J_m = \frac{1}{\sigma^2 + \gamma\nu^2}.$$

Shared model error therefore introduces an additional information ceiling. Nominal agent multiplicity or prompt diversity cannot remove error components common to all extractors.

6.4 Mutual information

If

$$\Theta \sim \mathcal{N}(0, \tau^2),$$

then

$$I(\Theta; R_1, \dots, R_m) = \frac{1}{2} \log(1 + \tau^2 J_m).$$

For independent extraction from one shared root,

$$I_{\text{shared}}(m) = \frac{1}{2} \log \left(1 + \frac{\tau^2 m}{\nu^2 + m\sigma^2} \right),$$

so

$$\lim_{m \rightarrow \infty} I_{\text{shared}}(m) = \frac{1}{2} \log \left(1 + \frac{\tau^2}{\sigma^2} \right).$$

The limit equals the information available from direct observation of the primitive source E .

7 Evidence Provenance

Theorem 1 implies that the information-generating structure cannot generally be recovered from report content alone. A richer interface can reveal relevant parts of that structure.

Let

$$G = (V, E_G)$$

be a directed acyclic provenance graph. Nodes can represent primitive observations, datasets, documents, transformations, retrieval operations, models, agents, and reports. An edge

$$u \rightarrow v$$

records that information from u was available in generating v .

For report R_i , let

$$\mathcal{R}(R_i)$$

denote its authenticated primitive root set.

7.1 Certifying zero novelty

Suppose the provenance model establishes

$$Z = f(R, U),$$

where

$$U \perp \Theta \mid R.$$

Then

$$\Theta \perp Z \mid R$$

and therefore Z is an epistemic Sybil extension.

Proposition 3 (Posterior invariance). *If*

$$I(\Theta; Z \mid R) = 0,$$

then

$$P(\Theta \mid R, Z) = P(\Theta \mid R)$$

almost surely.

Proof. Zero conditional mutual information implies the required conditional independence, yielding the posterior identity. $\square$

Under a sufficiently informative interface, ordinary Bayesian updating is therefore automatically invariant to certified zero-novelty extensions. The difficult problem is identifying the relevant dependence structure.

7.2 Model parameters as evidence

In generative AI systems, the phrase “no additional external evidence” is insufficient.

Suppose an LLM produces

$$Z = f(R, W, U),$$

where W denotes model parameters. Those parameters encode regularities acquired during training and can contain information about Θ that is absent from R . In general,

$$I(\Theta; W \mid R)$$

need not be zero.

A model instructed only to paraphrase a report can therefore fail to be a pure transformation channel if it corrects, supplements, or reinterprets the report using parametric knowledge.

To certify zero novelty one needs an assumption such as

$$\Theta \perp (W, U) \mid R,$$

or an explicit modeling convention under which the model’s parametric state is not treated as additional admissible evidence for the task.

Provenance must therefore characterize relevant information access, not merely external document retrieval.

7.3 Root disjointness is not statistical independence

Even when

$$\mathcal{R}(R_i) \cap \mathcal{R}(R_j) = \emptyset,$$

it need not follow that

$$R_i \perp R_j \mid \Theta.$$

Distinct datasets can sample the same population, independent sensors can share environmental noise, and apparently independent documents can depend on an unrecorded common source.

A provenance interface can establish explicit derivation structure. It cannot prove the absence of every latent common cause.

8 A Hierarchy of Information Interfaces

The preceding sections suggest a hierarchy rather than a binary distinction between having and not having provenance.

8.1 Report-only interface

Under $\mathcal{J}_0$, the mechanism observes report contents and ordinary report metadata but no information capable of distinguishing their evidential ancestry.

Theorem 1 establishes non-identifiability over any model class containing both clone and independent-corroboration structures.

8.2 Root interface

A stronger interface reveals authenticated root sets

$$\{\mathcal{R}(R_i)\}_{i=1}^n.$$

This identifies known shared ancestry but does not determine the strength of statistical dependence induced by that ancestry. Section 6 shows why counting roots is not generally sufficient.

8.3 Structural provenance interface

A richer interface exposes the derivation DAG or selected predicates over it. Such information can establish, for example, that a new report was generated only from already admitted reports.

It still does not identify unobserved statistical dependencies.

8.4 Probabilistic provenance interface

The strongest practical interface augments lineage with a generative dependence model, covariance structure, conditional-independence assumptions, or bounds on unknown dependencies.

At that point, the aggregation problem reduces to probabilistic inference under the specified model.

The design relation is therefore

$$\text{provenance structure} + \text{dependence assumptions} \longrightarrow \text{evidential contribution.}$$

Provenance alone should not be interpreted as evidential weight.

9 Conservative Aggregation Under Incomplete Dependence Information

Complete probabilistic provenance will rarely be available. Robust fusion methods become relevant precisely in this intermediate regime.

Suppose numerical estimates have known marginal covariance matrices but unknown cross-covariance. Treating them as independent can yield an overconfident posterior. Covariance Intersection instead chooses a fused information representation that remains consistent for the admissible unknown correlations.

This suggests three regimes:

$$\text{known independence} \longrightarrow \text{full accumulation,}$$

$$\text{known dependence} \longrightarrow \text{model-based aggregation,}$$

$$\text{unknown dependence} \longrightarrow \text{conservative aggregation.}$$

Overlapping Covariance Intersection shows that partial structural knowledge about shared correlation can improve upon fully correlation-agnostic fusion. Evidence provenance could provide analogous structural information in applications where uncertainty is represented probabilistically.

This paper does not propose a replacement for covariance-intersection methods. Its contribution is to formulate the informational invariance criterion and the side-information problem at a level that also covers reports whose representations are not conventional numerical state estimates.

10 Strategic Provenance

The previous sections assume that provenance information is correct.

In strategic systems this assumption is itself problematic.

Suppose participant i possesses report R_i , true private provenance G_i , and utility determined by the aggregate decision. It can instead declare $\hat{G}_i$. A truthful-provenance mechanism would seek conditions under which

$$\mathbb{E}[u_i(A(R, G))] \geq \mathbb{E}\left[u_i\left(A(R, \hat{G}_i, G_{-i})\right)\right]$$

for admissible provenance deviations.

If independent evidence receives greater marginal influence, an agent benefits from suppressing ancestry. A coalition can attempt to manufacture apparently separate roots. If a system rewards agreement, endogenous copying can create precisely the correlation that some information-elicitation mechanisms interpret as evidence of informativeness.

A complete mechanism would therefore need to address three distinct objects:

content reports,

provenance reports,

and

dependence-aware aggregation.

Truthful provenance remains an open mechanism-design problem.

11 Privacy-Preserving Provenance

Useful provenance can be confidential.

A journalist may wish to establish that two reports originate from separate sources without identifying either source. Firms may wish to demonstrate non-overlapping evidential ancestry without exposing proprietary datasets. AI agents may need to prove that retrieved evidence follows different derivation paths while keeping the underlying documents private.

This suggests cryptographic side-information interfaces in which participants commit to provenance structures and prove predicates such as

$$\mathcal{R}(R_i) \cap \mathcal{R}(R_j) = \emptyset$$

or

$$|\mathcal{R}(R_i) \cap \mathcal{R}(R_j)| \leq k$$

without disclosing the root identities themselves.

Origin-bound authority and content-addressed lineage already appear in recent agent-security and collective-inference systems, which makes such an interface technically less speculative than the underlying epistemological problem might initially suggest.

The limitation is important. Cryptography can establish properties of a committed provenance structure. It cannot establish that two real-world observations are statistically independent merely because their committed source identifiers differ.

Privacy-preserving provenance is therefore a mechanism for communicating side information, not a substitute for a probabilistic dependence model.

12 Simulation Study

The analytical model makes quantitative predictions that can be checked directly. We therefore conduct a controlled Monte Carlo study before considering experiments involving language models or human evidence.

12.1 Design

The data-generating process follows Section 6.

We set

$$\Theta \sim \mathcal{N}(0, \tau^2),$$

with baseline parameters

$$\tau^2 = \sigma^2 = \nu^2 = 1.$$

For primitive root j ,

$$E_j = \Theta + \varepsilon_j, \quad \varepsilon_j \sim \mathcal{N}(0, \sigma^2).$$

Reports assigned to root j satisfy

$$R_{ij} = E_j + \eta_{ij}, \quad \eta_{ij} \sim \mathcal{N}(0, \nu^2).$$

We vary total report count

$$n \in \{1, 2, 4, 8, 16, 32\}$$

and the number of independent primitive roots

$$k \in \{1, 2, 4, 8, 16, 32\}, \quad k \leq n.$$

Reports are allocated as evenly as possible across roots. Each (n, k) cell contains 60,000 Monte Carlo realizations, using a fixed random seed of 20260818.

Two aggregators are compared.

The dependence-aware Bayesian aggregator uses the correct shared-root covariance structure.

The naive aggregator treats all n reports as conditionally independent observations with variance

$$\sigma^2 + \nu^2.$$

No parameters are estimated from the simulated outcomes. The simulation therefore evaluates the consequences of the assumed information structures rather than fitting a model to produce the predicted behavior.

12.2 Metrics

We evaluate posterior RMSE, posterior standard deviation, nominal 95% credible-interval coverage, and logarithmic score.

Define the calibration ratio

$$C = \frac{\text{RMSE}}{\text{reported posterior standard deviation}}.$$

A correctly calibrated Gaussian posterior has $C \approx 1$. Values substantially greater than one indicate overconfidence.

The realized log-score improvement produced by admitting a report Z is

$$\Delta\ell = \log P(\Theta \mid R, Z) - \log P(\Theta \mid R).$$

When these probabilities are the true posteriors under the data-generating model,

$$\mathbb{E}[\Delta\ell] = I(\Theta; Z \mid R).$$

This identity provides an operational interpretation of the theoretical quantity $\Gamma(Z; R)$. Candidate aggregators are evaluated against this ideal, not assumed to satisfy the identity themselves.

12.3 Report multiplication with one evidence root

We first hold

$$k = 1$$

while increasing n .

The dependence-aware posterior remains calibrated throughout. Its empirical coverage stays near the nominal 95% level for every n .

The naive aggregator behaves very differently. Additional reports cause its reported posterior variance to shrink as if new independent evidence were arriving, while the common source error cannot be averaged away.

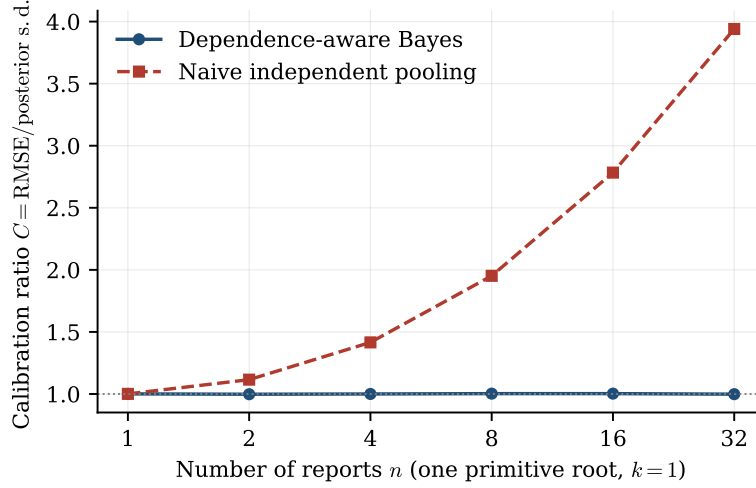


Figure 1: Calibration ratio as report count increases with one primitive evidence root. The dependence-aware posterior remains close to one; the independence-assuming posterior becomes progressively overconfident.

Reports n	Bayes coverage	Naive coverage	Bayes NLL	Naive NLL
1	0.949	0.949	1.217	1.217
2	0.950	0.921	1.159	1.192
4	0.949	0.833	1.128	1.378
8	0.950	0.686	1.099	1.999
16	0.950	0.521	1.091	3.673
32	0.950	0.381	1.077	7.273

At $n = 32$, the naive aggregator’s RMSE is approximately 3.94 times its own reported posterior standard deviation. A nominal 95% credible interval contains the true state only 38.1% of the time.

The additional reports have therefore produced substantially greater confidence while sharply degrading probabilistic performance.

12.4 Report count versus root count

Next fix

$$n = 16$$

and vary k .

The information contained in the reports under the correct model is:

Primitive roots k	$I(\Theta; R)$, nats
1	0.332
2	0.511
4	0.718
8	0.923
16	1.099

The naive aggregator assigns 1.099 nats of implied information in every case because it reacts only to $n = 16$, not to the evidential ancestry.

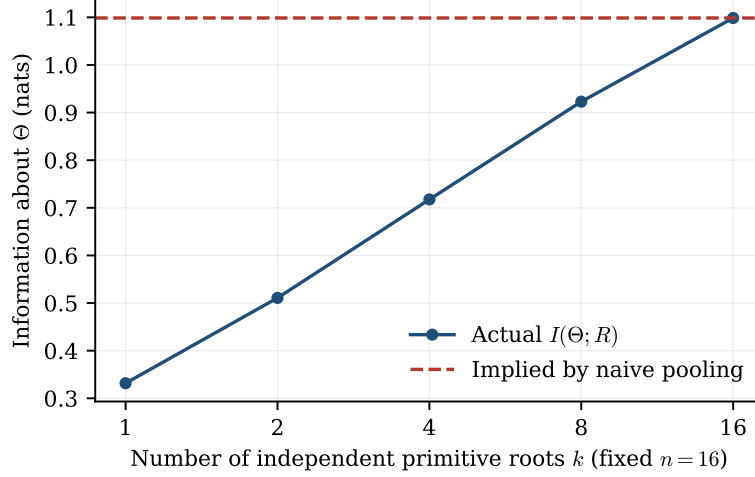


Figure 2: Mutual information at fixed $n = 16$ as the number of independent roots varies. Report count is constant; evidential information is not.

Thus the same number of reports ranges from 0.332 to 1.099 nats of actual information depending only on how many independent roots generated them.

12.5 Shared-root saturation

The analytical scaling law of Proposition 2 is also visible directly.

With one shared root,

$$J_m = \frac{m}{\nu^2 + m\sigma^2}$$

approaches the ceiling

$$\frac{1}{\sigma^2}.$$

With independent roots,

$$J_m^{\text{ind}} = \frac{m}{\sigma^2 + \nu^2},$$

so precision continues to grow linearly.

12.6 Log-score deterioration

Increasing report multiplicity with one fixed root does more than misstate uncertainty.

The naive negative log score rises from approximately 1.22 at $n = 1$ to 7.27 at $n = 32$. The dependence-aware Bayesian score instead improves slightly as repeated extraction removes part of the extraction noise.

12.7 Interpretation

The simulation should not be interpreted as empirical evidence that real LLM agents follow the Gaussian model. The dependence structure is imposed by construction.

Its role is narrower.

First, it numerically verifies the implementation of the analytical model.

Second, it shows that the difference between report count and evidence-root count can generate large calibration errors even in a minimal setting.

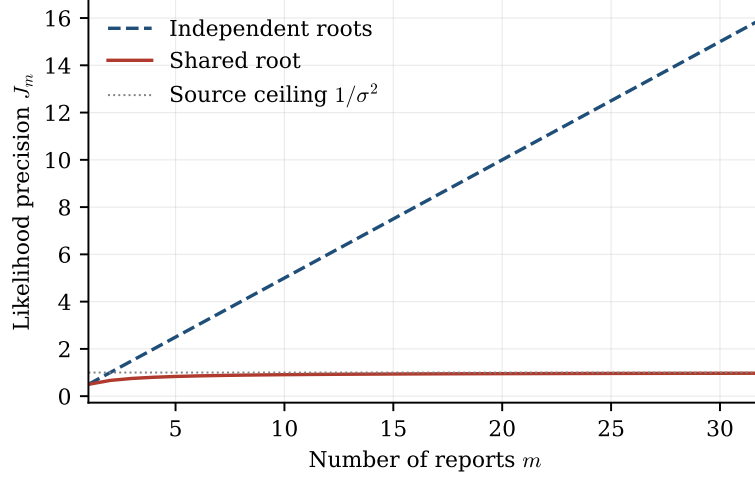


Figure 3: Likelihood precision under shared-root extraction and independent-root corroboration. Shared-root precision approaches a finite ceiling, whereas independent-root precision continues to accumulate.

Third, it provides quantitative hypotheses for a subsequent empirical experiment in which report similarity, evidence ancestry, agent diversity, and report count can be manipulated independently.

13 Empirical Predictions

The framework produces several falsifiable predictions beyond the controlled Gaussian setting.

13.1 Multiplicity inflation

Holding primitive evidence fixed while increasing the number of derived reports should increase confidence faster than predictive performance in aggregators that assume independence.

13.2 Shared-root saturation

When several agents independently extract information from a fixed noisy source, additional agents can improve inference, but improvement should exhibit diminishing returns toward a source-dependent ceiling.

The relevant empirical target is the existence and approximate level of the ceiling rather than direct estimation of the asymptotic $O(m^{-2})$ rate.

13.3 Provenance and propagation

In environments where one evidential root is retransmitted through multiple agents, an aggregator supplied with correct ancestry information should achieve better calibration and logarithmic score than a report-only aggregator receiving the same terminal reports.

13.4 Semantic similarity and evidential independence

A decisive experiment should manipulate semantic similarity independently from provenance:

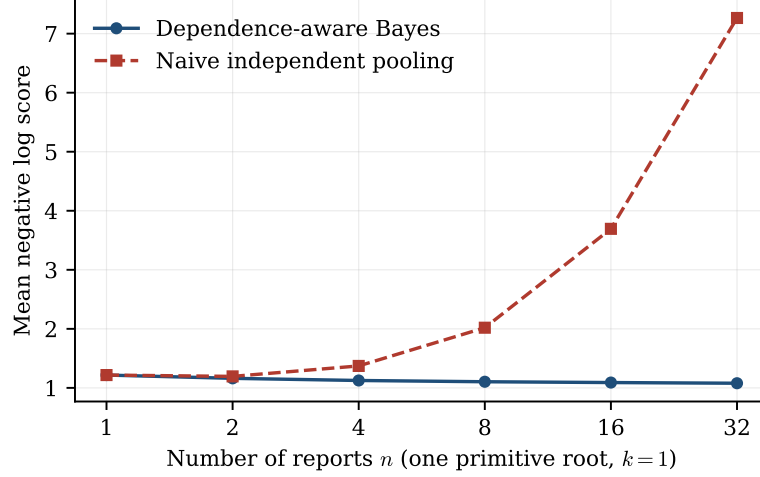


Figure 4: Mean negative log score under propagation from a single evidence root. Independence-assuming aggregation deteriorates rapidly as report multiplicity increases.

	Shared root	Independent roots
Similar reports	A	B
Dissimilar reports	C	D

A report-space deduplication mechanism risks false merging in cell B and false splitting in cell C.

A provenance-aware method should distinguish those cells when the experimental data-generating process guarantees the claimed ancestry.

This experiment directly operationalizes the non-identifiability problem in Theorem 1.

14 Discussion

Collective inference systems commonly count agents, documents, models, votes, or reports. None of these quantities is generally equivalent to the amount of independent evidence available about a latent state.

Many agents can carry one informational root. One agent can possess several independent observations. Reports that share a root can still contain partially distinct information. Reports with different root labels can remain statistically dependent because of latent common causes.

The quantity

$$I(\Theta; Z \mid R)$$

captures the relevant distinction at the probabilistic level. It measures the information about Θ that becomes available from Z after everything already represented by R is known.

This perspective also clarifies the relationship among several existing approaches.

Classical Sybil resistance protects identity systems against artificial participant multiplicity.

Clone robustness limits influence created by report duplication or near-duplication.

Data-incest methods prevent repeated use of information in distributed estimation.

Covariance Intersection handles numerical fusion when cross-correlation is unknown.

Correlation-neglect research studies the behavioral consequences of treating dependent information as independent.

Provenance systems expose derivation structure.

The epistemic Sybil problem lies at the intersection of these ideas because its protected resource is marginal information about a latent state rather than identities or representations.

The Gaussian model additionally shows that a binary duplicate/non-duplicate distinction is insufficient. For equicorrelated reports, the effective evidence count is

$$m_{\text{eff}} = \frac{m}{1 + \rho(m - 1)}.$$

As ρ varies continuously from zero toward one, the evidential structure moves continuously from independent corroboration toward effective replication.

The general problem is therefore not deduplication. It is dependence-aware aggregation under incomplete information about how reports were generated.

15 Limitations

Several limitations delimit the contribution.

First, conditional mutual information is defined relative to a probability distribution. In many real inference problems the joint distribution is unknown.

Second, provenance records known derivation, not every latent common cause.

Third, model parameters, human background knowledge, and other unobserved states can constitute evidence even when no additional document or sensor is consulted.

Fourth, provenance authenticity does not establish the truth of primitive evidence.

Fifth, primitive evidence roots can themselves be Sybil-manipulated unless their creation is constrained by domain-specific authentication, resource, hardware, institutional, or identity assumptions.

Sixth, the Gaussian model is deliberately stylized. Its purpose is to distinguish exact duplication, shared-root extraction, and independent corroboration, and to derive a closed-form dependence correction. It is not proposed as a universal model of natural-language evidence.

Seventh, the Monte Carlo study is a model validation exercise rather than empirical confirmation of the framework in deployed collective systems.

Finally, this paper does not provide a universal estimator of

$$I(\Theta; Z \mid R).$$

Without structural assumptions or side information, Theorem 1 shows why no universal report-only solution should be expected.

16 Conclusion

When producing reports becomes inexpensive, report multiplicity ceases to be a reliable proxy for evidential multiplicity.

This paper formalizes the resulting problem through conditional mutual information. An additional report Z is an epistemic Sybil extension relative to admitted reports R when

$$I(\Theta; Z \mid R) = 0.$$

The definition is independent of agent identity and report representation.

Three main conclusions follow.

First, the relevant information structure cannot generally be identified from reports alone. Two identical reports can represent either pure replication or genuinely independent corroboration and require different Bayesian treatment.

Second, descendants of fixed evidence cannot collectively contain more information about the latent state than their evidential ancestors. Report multiplication cannot mint information.

Third, common ancestry does not imply complete redundancy. Multiple noisy extractions from one source can add genuine information, but that information exhibits diminishing returns toward a source-level ceiling. Under Gaussian equicorrelation, the corresponding evidence discount is exactly

$$\frac{1}{1 + \rho(m - 1)}.$$

The simulation study shows that violating this structure can generate severe overconfidence. With 32 reports derived from one primitive root, independence-assuming aggregation reduces nominal 95% posterior coverage to 38.1%, even though a dependence-aware posterior remains correctly calibrated.

These results shift the central question from how many reports agree to how much conditionally novel evidence the reports collectively contain.

The next theoretical problem is consequently precise:

«What is the weakest side-information interface under which an aggregator can recover, or approximate with bounded loss, the posterior available under the full information-generating structure?»

A solution would characterize the minimum provenance or dependence information required for reliable collective inference. Truthful provenance, privacy-preserving provenance, and aggregation under partially known dependence then become connected subproblems rather than separate engineering concerns.

Epistemic Sybil resistance is therefore not a proposal to count sources more carefully. It is a proposal to make marginal information, rather than report multiplicity, the invariant of collective inference.

References

- Berriaud, D., and Wattenhofer, R. (2026). Breaking the Illusion of Artificial Consensus: Clone-Robust Weighting for Arbitrary Metric Spaces. arXiv:2602.24024.
- Blackwell, D. (1953). Equivalent comparisons of experiments. *Annals of Mathematical Statistics*, 24(2), 265–272.
- Burnat, F. A. D., and Davidson, B. I. (2026). Quotient Semivalues for False-Name-Resistant Data Attribution. arXiv:2605.07663.
- Cheney, J., Ahmed, A., and Acar, U. A. (2007). Provenance as Dependency Analysis. arXiv:0708.2173.
- Clemen, R. T., and Winkler, R. L. (1999). Combining probability distributions from experts in risk analysis. *Risk Analysis*, 19(2), 187–203.
- Clemen, R. T., and Winkler, R. L. (2007). Aggregating probability distributions. In W. Edwards, R. F. Miles Jr., and D. von Winterfeldt (Eds.), *Advances in Decision Analysis*. Cambridge University Press.
- DeMarzo, P. M., Vayanos, D., and Zwiebel, J. (2003). Persuasion bias, social influence, and unidimensional opinions. *Quarterly Journal of Economics*, 118(3), 909–968.
- Douceur, J. R. (2002). The Sybil attack. In *Peer-to-Peer Systems, IPTPS 2002, Lecture Notes in Computer Science 2429*, 251–260.

- Enke, B., and Zimmermann, F. (2019). Correlation neglect in belief formation. *Review of Economic Studies*, 86(1), 313–332.
- Golub, B., and Jackson, M. O. (2010). Naïve learning in social networks and the wisdom of crowds. *American Economic Journal: Microeconomics*, 2(1), 112–149.
- Julier, S. J., and Uhlmann, J. K. (1997). A non-divergent estimation algorithm in the presence of unknown correlations. *Proceedings of the American Control Conference*, 2369–2373.
- Kong, Y., and Schoenebeck, G. (2016). An Information Theoretic Framework for Designing Information Elicitation Mechanisms That Reward Truth-telling. *arXiv:1605.01021*.
- Lee, D., and Kim, J. (2026). A Failure-Mode Benchmark for Polymorphic Sybil Poisoning in RAG. *arXiv:2607.03739*.
- Louck, Y. (2026). Securing LLM-Agent Long-Term Memory Against Poisoning: Non-Malleable, Origin-Bound Authority with Machine-Checked Guarantees. *arXiv:2606.24322*.
- McLaughlin, S. P., Evans, R. J., and Krishnamurthy, V. (2003). Data incest removal in a survivable estimation fusion architecture. *Proceedings of the Sixth International Conference of Information Fusion*.
- Pedroso, L., Batista, P., and Heemels, W. P. M. H. (2026). Overlapping Covariance Intersection: Fusion with Partial Structural Knowledge of Correlation from Multiple Sources. *arXiv:2603.16768*.
- Williams, P. L., and Beer, R. D. (2010). Nonnegative Decomposition of Multivariate Information. *arXiv:1004.2515*.
- Winkler, R. L. (1981). Combining probability distributions from dependent information sources. *Management Science*, 27(4), 479–488.
- Wu, Z. (2026). Split the Labor: Separating Evidence Interpretation from Decision Aggregation. *arXiv:2608.14509*.
- Xu, H. (2026). Mesh Inference: A Formal Model of Collective Inference Without a Center. *arXiv:2606.19537*.
- Yan, C., Yue, Z., Zhao, F., Lin, E., Jia, L., Tong, H., Lyu, M., Sun, C., and Zeng, Y. (2026). When Truth Is Distributed: Misinformation Derails Collective Fact Recovery in LLM-Based Multi-Agent Systems. *arXiv:2608.03421*.